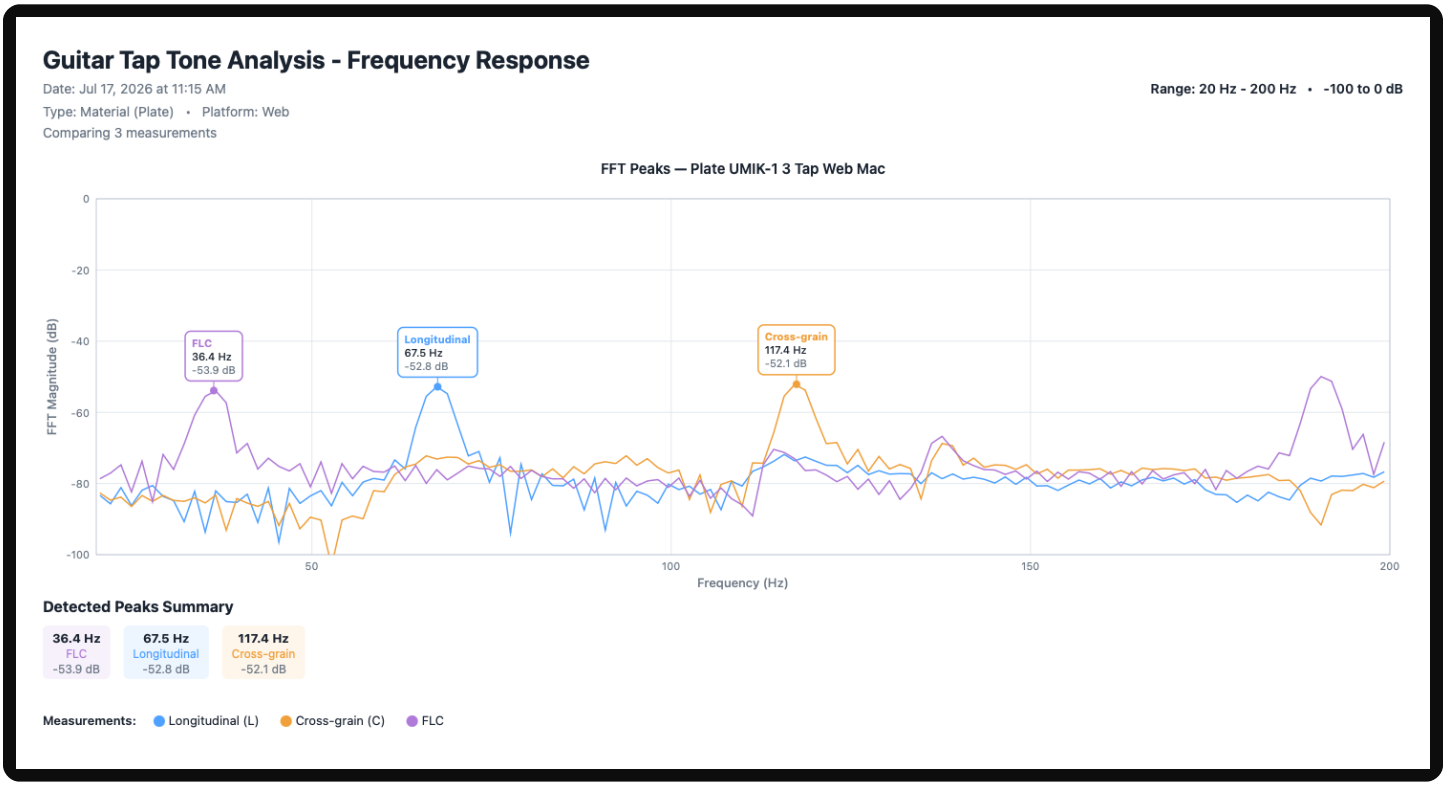
GuitarTap

Tap Tone Analysis Report

Jul 17, 2026 at 11:15 AM

Measurement Name: Plate UMIK-1 3 Tap Web Mac
Type: Material (Plate)
Frequency Range: 20.0 Hz – 200.0 Hz
Microphone: Umik-1 Gain: 18dB · calibrated (7108913)

Frequency Spectrum



Detected Peaks

Frequency	Magnitude	Note	Q Factor	Role
36.4 Hz	-53.9 dB	D1	8.3	FLC (Diagonal)
67.5 Hz	-52.8 dB	C#2	11.5	Longitudinal (L)
117.4 Hz	-52.1 dB	A#2	26.7	Cross-grain (C)

Gore Target Thickness

2.78 mm

Body: 490 × 368 mm · f_vs = 60 (Classical Top)
GLC (Shear Modulus): 0.360 GPa

Plate Properties

Sample Dimensions

Length: **557.5 mm** | Width: **220.5 mm** | Thickness: **4.85 mm**

Mass: **208.0 g** | Density: **0.349 g/cm³**

fL: **67.5 Hz** | fC: **117.4 Hz** | fLC: **36.4 Hz**

Speed of Sound (L): **4209 m/s**

Speed of Sound (C): **1146 m/s**

Young's Modulus (L): **6.18 GPa**

Young's Modulus (C): **0.46 GPa**

GLC (Shear Modulus): **0.360 GPa**

Specific Modulus (L): **17.7** (Fair)

Specific Modulus (C): **1.3** (Very Good)

Radiation Ratio (L): **12.1**

Radiation Ratio (C): **3.3**

Cross/Long Ratio: **0.074**
typical: 0.04–0.08

Long/Cross Ratio: **13.5**
typical: 12–25

Overall Quality: **Good**

Three-Tap Measurement Process:

- **1. Longitudinal (L) Tap**
  Hold plate at 22% from one end along the length, near one long edge (not at the width node). Tap center.
- **2. Cross-grain (C) Tap**
  Rotate 90°. Hold plate at 22% from one end along the width, near one short edge (not at the length node). Tap center.
- **3. FLC (Diagonal) Tap**
  Hold plate at the midpoint of one long edge. Tap near the opposite corner (~22% from both the end and the side). Measures shear stiffness.

The strongest peak from each tap is auto-selected.